Weiyong Xu

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Research Interests

Topics: Learning and memory, audiovisual integration, reading acquisition, memory reorganisation and brain-body interactions

Methods: MEG, EEG, physiological recordings, multivariate pattern analysis, representational similarity analysis, computational modelling and one-photon calcium imaging

Education

PhD **University of Jyväskylä**, Psychology — Finland
- Department: Department of Psychology — 2015 – 2020
- Thesis: Brain activity changes related to learning of audiovisual associations in reading

MS **Xi'an Jiaotong University**, Biomedical Engineering — China
- Department: Institute of Biomedical Engineering — 2013 – 2015
- Thesis: Functional connectivity and complex network analysis of the aging brain

BS **Xi'an Jiaotong University**, Biomedical Engineering — China
- Department: School of Life Science and Technology — 2009 – 2013
- Thesis: EEG functional connectivity of vascular dementia patients in an oddball task

Experience

Department of Psychology, University of Jyväskylä, Postdoctoral Researcher — Finland
Research Council of Finland-funded Trajectory project — Nov 2026 – Oct 2028 — 2 years
- PI: Prof. Tiina Parviainen. Investigating cortical development and neural mechanisms supporting cognition and learning in children.

Department of Psychology, University of Jyväskylä, Postdoctoral Researcher — Finland
Business Finland-funded DIGIMIND project: Psychophysiological measurement and monitoring as a digital indicator of mental state — Jan 2026 – Oct 2026 — 10 months
- Project leader: Prof. Tiina Parviainen.

Department of Psychology, University of Jyväskylä, Postdoctoral Researcher — Finland
University of Jyväskylä profiling action: Understanding learning variability, support, and change (ULearn) — Sept 2025 – Dec 2025 — 4 months

Department of Psychology, University of Jyväskylä, Postdoctoral Researcher — Finland
Project: The reorganization of memory representations during rest and sleep — Sept 2022 – Aug 2025 — 3 years
- Independent Principal Investigator of the Research Council of Finland-funded postdoctoral project (EUR 234,782), responsible for project management, data collection and dissemination.
- Investigated memory reorganisation during rest and sleep using MEG/EEG, behavioural measures and one-photon calcium imaging, and developed a person-centred event-learning paradigm.

Department of Psychology, University of Jyväskylä, Postdoctoral Researcher — Finland
Project: Bodily rhythms and learning — Sept 2020 – Aug 2022
2 years

- Investigated the interaction between physiological rhythms (such as heartbeat and breathing) and learning processes using behavioral and brain measures (M/EEG) in the Research Council of Finland project "Optimizing Learning: Synchrony of the Brain and Body as a Tool?" (Dr. Miriam Nokia's group).
- Set up new calcium imaging systems and developed a data analysis pipeline for the Research Council of Finland project "Dentate gyrus-the gateway to memory?" (Dr. Markku Penttonen).

Department of Psychology, University of Jyväskylä, Doctoral Student — Finland
Project: Brain activity changes related to learning of audiovisual associations in reading — Sept 2015 – Sept 2020
5 years 1 month

- Doctoral Researcher (2019-09 – 2020-09)
- Early Stage Researcher (2015-09 – 2019-09), ChildBrain Project, Horizon2020 Marie Skłodowska-Curie Action (MSCA) Innovative Training Network (ITN)

Research Visits

Visiting Researcher — Finland
Aalto University — Feb 2025 – Mar 2025

- Department: Department of Neuroscience and Biomedical Engineering
- Host: Prof. Lauri Parkkonen

Visiting Researcher — Japan
Tohoku University — May 2025 – May 2025

- Host: Prof. Hyeonjeong Jeong

Visiting Student — Netherlands
Donders Institute for Brain, Cognition and Behaviour — July 2016 – July 2016

- Host: Prof. Robert Oostenveld
- Learn MEG data analysis methods

Visiting Student — UK
Aston Brain Center — Feb 2018 – Feb 2018

- Host: Prof. Joel B. Talcott and Dr. Caroline Witton
- Learn child MEG and MRI measurements and data analysis

Funding

The reorganization of memory representations during rest and sleep — Sept 2022 – Aug 2025
Research Council of Finland postdoctoral project

- Role: Principal Investigator
- Amount: 234,782 EUR

Research Software

rsa-nd — 2026
Open-source Python package extending MNE-Python and mne-rsa for representational similarity analysis across EEG/MEG sensors or brain regions, time, frequency and oscillatory phase.

- DOI: 10.5281/zenodo.24088896

Calcium Imaging Cell Sorter — 2026
Open-source Python tool for reviewing cell images and activity traces in Inscopix calcium-imaging recordings and documenting cell-selection decisions.

- DOI: 10.5281/zenodo.24089476

Preprints

Affective representations and brain-body dynamics during emotional video viewing — Oct 2026
Xu, W., Ronkainen, P., Parviainen, T.
10.64898/2026.10.28.762567 (bioRxiv [Preprint])

Many Pipelines, One Dataset: Analytic Flexibility and Its Impact in EEG Research — 2026
Cesnaite, E., et al. (including **Xu, W.**)
10.31222/osf.io/wvx48_v2 (MetaArXiv [Preprint])

The EEGManyPipelines Dataset: Metascientific Data on 168 Independent Analyses of a Single EEG Dataset — 2025
Cesnaite, E., et al. (including **Xu, W.**)
10.31222/osf.io/c4xwg_v1 (MetaArXiv [Preprint])

Invited Talks

Neural correlates of retrospective memory confidence during face–name associative learning — Sendai, Japan 2025
26th Human Brain Science Seminar, Tohoku University

Oral Presentations

Representational Similarity Analysis for Cognitively Grounded Interpretation of MEG Dynamics — Jyväskylä, Finland Sept 2026
CuttingGardens

Representational similarity analysis for cognitively grounded interpretation of magnetoencephalographic dynamics — Beijing, China Aug 2026
BIOMAG 2026 (symposium talk)

Neural correlates of retrospective memory confidence during face–name associative learning — Äkäslompolo, Finland March–April 2026
The Learning Brain Across the Lifespan: Neural Resources, Imaging Indicators and Relevance for Cognitive Competence, Learning and Plasticity Meeting 2026

From Novice to Expert: Different Levels of Learning in the Brain Revealed by MEG — Jyväskylä, Finland Jan 2026
Jyväskylä Brain & Mind Annual Seminar

Neural correlates of retrospective memory confidence during face–name associative learning — Porto, Portugal 2025
International Conference on Cognitive Neuroscience (ICON 2025)

The reorganization of memory representations during rest and sleep — Espoo, Finland 2025
Neuroscience and Biomedical Engineering Seminar, Aalto University

Neural correlates of retrospective memory confidence during face–name associative learning — Online 2025
Staresina Lab Meeting

Differential brain processes of newly learned and overlearned audiovisual associations — Jyväskylä, Finland 2024
Jyväskylä Brain & Mind Annual Seminar 2024

Neural correlates of retrospective memory confidence during face–name associative learning — Jyväskylä, Finland 2023
MEG Nord 2023

Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children — Leuven, Belgium, 2018
ChildBrain Conference 2018

Audiovisual processing of Chinese characters elicits integration and congruency effects in MEG — Stockholm, Sweden, 2018
MEG Nord 2018

Poster Presentations

Differential brain processes of newly-learned and overlearned audiovisual associations — Jyväskylä, Finland, 2024
Brains and Bodies in Social Interaction, Learning, and Wellbeing Conference 2024

Differential brain processes of newly-learned and overlearned audiovisual associations — Vienna, Austria, 2024
FENS Forum 2024

Changes in brain activity during learning of grapheme–phoneme associations — Helsinki, Finland, 2019
Society for the Neurobiology of Language (SNL 2019)

Changes in brain activity during learning of grapheme–phoneme associations — Jyväskylä, Finland, 2019
MEG Nord 2019

Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children — San Diego, USA, 2018
Society for Neuroscience (SfN 2018)

Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children — Jyväskylä, Finland, 2018
International Summer School "Life after PhD"

Long-term learning effect on audio-visual processing of Chinese characters — Taipei, Taiwan, 2017
International Conference on Cognitive Science (ICCS 2017)

Development of audio-visual integration in children — Jyväskylä, Finland, 2017
International Summer School "Understanding Learning in the Brain"

Audio-visual processing of written characters and speech sounds — Aarhus, Denmark, 2017
MEG Nord 2017

Audio-visual processing of Chinese characters and speech sounds — Seoul, South Korea, 2016
Biomag 2016 (20th International Conference on Biomagnetism)

Workshops, Seminars, and Training

Cutting EEGX 2024 — Online, 2024
Workshop

Brain & Mind Symposium 2023 — Espoo, Finland, 2023
Workshop

eSleep Europe 2023 — Online, 2023
Workshop

Brain and Language 2021: Virtual Conference Methods Workshop — Jyväskylä, Finland, 2021
Workshop

6th Annual Research Seminar of CIBR — Jyväskylä, Finland
Workshop — 2018

ChildBrain 6th Workshop, Donders Institute — Nijmegen, Netherlands
Workshop — 2018

5th Annual Research Seminar of CIBR — Jyväskylä, Finland
Workshop — 2017

ChildBrain 5th Workshop, Aston University — Birmingham, UK
Workshop — 2017

Nordic & Baltic Doctoral Programmes in Psychology Joint Symposium — Tartu, Estonia
Workshop — 2017

ChildBrain 4th Workshop, Donders Institute — Nijmegen, Netherlands
Workshop — 2017

4th Annual Research Seminar of CIBR — Jyväskylä, Finland
Workshop — 2016

ChildBrain 3rd Workshop — Leuven, Belgium
Workshop — 2016

Toolkit of Cognitive Neuroscience: Advanced data analysis and source modelling of EEG/MEG — Nijmegen, Netherlands
Workshop — 2016

Winter School on Neural Mechanisms of Learning — Jyväskylä, Finland
Workshop — 2016

Elekta Neuromag Triux MEG System Start Program, CIBR — Jyväskylä, Finland
Workshop — 2016

ForLearning International Symposium on Learning and Learning Difficulties — Jyväskylä, Finland
Workshop — 2015

ChildBrain 1st Workshop (Kick-off) — Münster, Germany
Workshop — 2015

MEG Driving License — University of Jyväskylä, Finland
Workshop — 2015

Teaching

Examiner of one master's thesis — 2026
University of Jyväskylä

PSYS2102 Master's thesis ending seminar — 2023
instructor
- University of Jyväskylä

CNSJ1202 Designing cognitive neuroscience research — 2022
instructor
- University of Jyväskylä

CIBA120 EEG data analysis Demo using MNE Python 2022
demo instructor
- University of Jyväskylä

CIPA113 Bilingual Development 2021
essays grading
- University of Jyväskylä

CIPA112 Psychology of Language 2021
essays grading
- University of Jyväskylä

CIPA213 Developmental Language Related Disorders 2019
instructor
- University of Jyväskylä

CIPA100 Psychology Research at JYU 2016 – 2019
essays grading
- University of Jyväskylä

PSYS123 Neuropsychology 3B MEG demonstrations 2017
instructor
- University of Jyväskylä

Introduction to scripting experiments with Presentation 2016
instructor
- University of Jyväskylä

Awards

Winner of the best poster competition 2017
Annual Research Seminar of Centre for Interdisciplinary Brain Research (CIBR)

Siyuan Student Scholarship 2010 – 2013
Xi'an Jiaotong University

Service and Outreach

Conference organisation: Local co-organiser of CuttingGardens, Jyväskylä, and chair of "Making Sense of Brain Signals: Advances in M/EEG Methods", 22 September 2026.

Membership: Brain Research Society of Finland (BRSF), Federation of European Neuroscience Societies (FENS), International Brain Research Organisation (IBRO), Society for the Neurobiology of Language (SNL), Society for Neuroscience (SfN)

Ad Hoc Reviewer: NeuroImage, Frontiers in Psychology, Journal of Neuroscience, Child Development, Journal of Neurolinguistics, Computer Methods and Programs in Biomedicine, Acta Psychologica, npj Science of Learning, Journal of Open Source Software, Neuropsychologia

Supervision

Doctoral: Co-supervision of 4 doctoral researchers: 2 graduated in 2026 and 2 ongoing.

Master's: Supervision of 5 master's theses: 4 completed and 1 ongoing (Jukka Lopmeri).

Bachelor's: Supervision of 3 bachelor's theses.

Skills

Languages: Chinese (Mother tongue), English (Proficient)

Programming: Python, MATLAB, R

Neuroimaging: MEG analysis (MNE-Python, FieldTrip), EEG analysis, Calcium imaging analysis, Machine learning for neuroimaging

Tools: Git/GitHub, Linux/Unix

Scientific and Societal Impact

- Contributing analyst and coauthor in EEGManyPipelines, an international comparison of 168 teams' independent EEG analyses; contributed analysis results, scripts and methodological documentation.
- Developed and shared analysis code on GitHub for audiovisual integration (AVTest), memory confidence (FaceName-Brain), icon perception (IconMind) and speech-sound responses (JLD26).
- Coauthored research datasets released through JYX support reuse of learning experiments and their associated recordings; full dataset titles and DOIs are listed under Publications.
- Outreach: Organised activities for Researchers' Night at JYU (2016, 2017, 2020–2022 and 2024–2026).
- Tours: MEG lab tours for Finnish high school students, Bachelor and Master level students (2017)
- Tours: MEG lab tours for the Finnish Psychology Students' Association (2017)
- Events: Brain awareness week event "Developing Brain" at JYU (Poster presentation) (2016)

Publications

Cardiorespiratory rhythm and learning — Accepted for publication
Nokia, M. S., Lensu, S., Santhana Gopalan, P. R., **Xu, W.**
(Philosophical Transactions of the Royal Society B: Biological Sciences)

Cortical dynamics of icon perception: effects of concreteness and attractiveness — June 2026
Zheng, J., **Xu, W.**, Silvennoinen, J., Cong, F., Parviainen, T., Kujala, T.
10.1093/cercor/bhag075 (Cerebral Cortex)

Development of brain responses to speech sounds and associations with reading and reading related skills in adults with childhood dyslexia — Apr 2026
Sorsa, A., Tretow, A., **Xu, W.**, Gerike, G., Lohvansuu, K., Hämäläinen, J. A., Leppänen, P. H. T.
10.1016/j.ijpsycho.2026.113339 (International Journal of Psychophysiology)

Magnetoencephalography Source Connectivity Pipeline Reveals Long-Range Connectivity in Audio-Visual Integration — Apr 2026
Bornfleth, H., Cho, J.-H., Rusiniak, M., Kornwebel, O., **Xu, W.**, Hämäläinen, J., Spangler, R.
10.1111/ejn.70470 (European Journal of Neuroscience)

Neural signatures of automatic letter-speech sound integration in literate adults — Nov 2025
Xu, W., Li, X., Kolozsvari, O., Sorsa, A., Nokia, M., Hämäläinen, J.
10.1162/imag.a.1021 (Imaging Neuroscience)

Two-Day Artificial Orthography Learning MEG Dataset: From Grapheme to Meaning — Oct 2025
Xu, W., Kolozsvari, O., Hämäläinen, J., Oostenveld, R.
10.17011/jyx/dataset/106229 (JYX, University of Jyväskylä [Dataset])

Dynamic changes in electrical brain activity during passive learning of foreign speech sound discrimination — July 2025
Li, Q., Kurkela, J. L. O., Lohvansuu, K., Hämäläinen, J. A., Li, X., **Xu, W.**, Astikainen, P.
10.1093/cercor/bhaf181 (Cerebral Cortex)

Optimizing learning - synchrony of the brain and body as a tool? Nov 2024
Nokia, M., **Xu, W.**
10.17011/jyx/dataset/98410 (JYX, University of Jyväskylä [Dataset])

The cardiac cycle modulates learning-related interoception Aug 2024
Nokia, M. S., **Xu, W.**, Wikgren, J.
10.1016/j.tics.2024.05.007 (Trends in Cognitive Sciences)

Neural correlates of retrospective memory confidence during face-name associative learning May 2024
Xu, W., Li, X., Parviainen, T., Nokia, M.
10.1093/cercor/bhae194 (Cerebral Cortex)

Cardiorespiratory rhythm-contingent trace eyeblink conditioning in elderly adults May 2024
Santhana Gopalan, P. R., **Xu, W.**, Waselius, T., Wikgren, J., Penttonen, M., Nokia, M. S.
10.1152/jn.00356.2023 (Journal of Neurophysiology)

Cardiac cycle and respiration phase affect responses to the conditioned stimulus in young adults trained in trace eyeblink conditioning Mar 2022
Waselius, T., **Xu, W.**, Sparre, J. I., Penttonen, M., Nokia, M. S.
10.1152/jn.00298.2021 (Journal of Neurophysiology)

Coherence Between Brain Activation and Speech Envelope at Word and Sentence Levels Showed Age-Related Differences in Low Frequency Bands May 2021
Kolozsvári, O. B., **Xu, W.**, Gerike, G., Parviainen, T., Nieminen, L., Noiray, A., Hämäläinen, J. A.
10.1162/nol_a_00033 (Neurobiology of Language)

Rapid changes in brain activity during learning of grapheme-phoneme associations in adults Oct 2020
Xu, W., Kolozsvari, O. B., Oostenveld, R., Hämäläinen, J. A.
10.1016/j.neuroimage.2020.117058 (NeuroImage)

Neural Responses to Musical Rhythm in Chinese Children With Reading Difficulties June 2020
Chiang, C.-H., Hämäläinen, J., **Xu, W.**, Wang, H.-L.
10.3389/fpsyg.2020.01013 (Frontiers in Psychology)

Context-dependent minimisation of prediction errors involves temporal-frontal activation Feb 2020
Hsu, Y.-F., **Xu, W.**, Parviainen, T., Hämäläinen, J. A.
10.1016/j.neuroimage.2019.116355 (NeuroImage)

Brain activity changes related to learning of audiovisual associations in reading 2020
Xu, W.
urn.fi/URN:ISBN:978-951-39-8220-1 (JYU Dissertations 249, University of Jyväskylä [Doctoral dissertation])

Top-Down Predictions of Familiarity and Congruency in Audio-Visual Speech Perception at Neural Level July 2019
Kolozsvári, O. B., **Xu, W.**, Leppänen, P. H. T., Hämäläinen, J. A.
10.3389/fnhum.2019.00243 (Frontiers in Human Neuroscience)

Audiovisual Processing of Chinese Characters Elicits Suppression and Congruency Effects in MEG Feb 2019
Xu, W., Kolozsvári, O. B., Oostenveld, R., Leppänen, P. H. T., Hämäläinen, J. A.
10.3389/fnhum.2019.00018 (Frontiers in Human Neuroscience)

Aging-related changes of EEG synchronization during a visual working memory task Dec 2018
Teng, C., Cheng, Y., Wang, C., Ren, Y., **Xu, W.**, Xu, J.
10.1007/s11571-018-9500-6 (Cognitive Neurodynamics)

Brain Responses to Letters and Speech Sounds and Their Correlations With Cognitive Skills Related to Reading in Children Aug 2018
Xu, W., Kolozsvari, O. B., Monto, S. P., Hämäläinen, J. A.
10.3389/fnhum.2018.00304 (Frontiers in Human Neuroscience)

Assay of melamine in milk products with a pH-mediated stacking technique in capillary electrophoresis Mar 2014
Kong, Y., Yuan, J., Wang, Z., Hou, Z., Yu, J., Jin, H., Xie, M., Chang, N., **Xu, W.**, Zhao, Y., Tang, Y.
10.1002/jssc.201300603 (Journal of Separation Science)